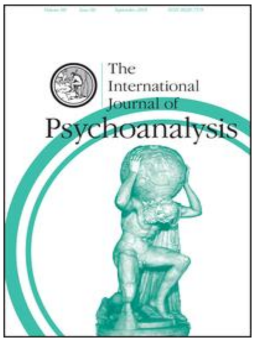

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Contemporary Conversations

Further evidence for the case against neuropsychoanalysis: How Yovell, Solms, and Fotopoulou's response to our critique confirms the irrelevance and harmfulness to psychoanalysis of the contemporary neuroscientific trend[1]

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In their paper "The case for neuropsychoanalysis" Yovell, Solms, and Fotopoulou (2015) respond to our critique of neuropsychoanalysis (Blass & Carmeli, 2007), setting forth evidence and arguments which, they claim, demonstrate why neuroscience is relevant and important for psychoanalysis and hence why dialogue between the fields is necessary. In the present paper we carefully examine their evidence and arguments and demonstrate how and why their claim is completely mistaken. In fact, Yovell, Solms, and Fotopoulou's paper only confirms our position on the irrelevance and harmfulness to psychoanalysis of the contemporary neuroscientific trend. We show how this trend perverts the essential nature of psychoanalysis and of how it is practiced. The clinical impact and its detrimental nature is highlighted by discussion of clinical material presented by Yovell et al (2015). In the light of this we argue that the debate over neuropsychoanalysis should be of interest to all psychoanalysts, not only those concerned with biology or interdisciplinary dialogue.

Keywords: biologism, biological reductionism, essence of psychoanalysis, meaning, meaningfulness, neuropsychoanalysis, neuroscience, psychoanalysis, scientific status, validation of psychoanalysis

We would like to start by stating that when we first received the invitation to respond to the paper by Yovell, Solms, and Fotopoulou (2015), 'The case for neuropsychoanalysis', we very much looked forward to the task. In our 2007 paper and in others (e.g., Carmeli and Blass, 2013) we argued that not only are neuroscientific findings completely irrelevant to psychoanalytic theory and practice, but that to wrongly believe that they are, as neuropsychoanalysts do, is very harmful to psychoanalysis. Neuropsychoanalysts took note of our work, but only cited it dismissively, without taking direct issue with the arguments and evidence that we put forth. And so we hoped that Yovell, Solms, and Fotopoulou's paper would provide serious discus-

[1]We are very grateful to Avi Kenan for his helpful comments on an earlier draft of this paper.

sion of our ideas, which in turn would enrich our thinking on these issues, perhaps change it, or at least refine it. We were especially pleased that the paper was written by authors so central to the field of neuropsychoanalysis, allowing us to regard their views as a representative approach to our critique.

After reading their paper we were sorry to realize that our hopes were not to be fulfilled. Ultimately, Yovell, Solms, and Fotopoulou's paper only confirm our earlier conclusions. What the authors present as contributions of neuroscientific knowledge to psychoanalysis, in our view, are either irrelevant to the field or harmful to it. Moreover, often what they call neuroscientific findings are actually general psychological ones. In addition, the arguments that we had put forth against neuropsychoanalysis are misrepresented in very fundamental ways and so it is only with a strawman that Yovell, Solms, and Fotopoulou take issue.

Despite our disappointment we think that the present exchange may help clarify our critique and explain why it should be of concern not only to those involved in the field of neuropsychoanalysis, but rather to all those who wish to better understand and preserve the essential nature of psychoanalysis, what uniquely defines it and makes it such a valuable practice. As may be seen from the following comments on Yovell, Solms, and Fotopoulou's paper, it is that which neuropsychoanalysis threatens.

In our response to their paper we will stay focused on the issue at hand – the arguments for neuropsychoanalysis as presented by Yovell, Solms, and Fotopoulou, our objections to them, and the nature of our differences, as perceived by them and by us. A large part of their paper is an account of the development of the field of neuropsychoanalysis. It has already appeared in various versions in neuropsychoanalytic publications, includes references to the Freudian origins of neuropsychoanalysis, to affirmations of its proponents that they are definitely *not* reductionistic, and to the interest neuropsychoanalysis has aroused in the psychoanalytic literature and among authoritative analytic authors. It also refers to the fact that the field has now evolved beyond some earlier, weaker versions of neuropsychoanalysis, which according to Yovell, Solms, and Fotopoulou are indeed worthy of critique and sings the praise of contemporary neuropsychoanalysis and its contributions. The presentation of this account and the massive bibliography that accompanies it may give the reader the sense that neuropsychoanalysis is a well-established and credible discipline and acquaint the reader with positive claims that have been put forth regarding the contribution of neuroscience to psychoanalysis. However, it does nothing to substantiate these claims, and offers no evidence or arguments regarding the value of neuroscience to psychoanalysis.

In what follows we take issue only with Yovell, Solms, and Fotopoulou's evidence and arguments. We first briefly summarize them and then we offer our response.

Yovell, Solms, and Fotopoulou's arguments for the value of neuropsychoanalysis

Some specific arguments and evidence in support of neuropsychoanalysis begin to appear somewhere in the middle of the section on Dual-Aspect Monism and involve four main steps which we summarize below.

1) The theoretical argument: Psychoanalysis and neuroscience study the same entity so evidently the knowledge of one contributes to the other

The theoretical argument runs as follows:

Psychoanalysis and neuroscience are two ways of studying the same thing, one subjective and the other objective.

> Since, however, both observation methods study the same entity (the so-called 'mindbrain'), and since, inherently, neither is sufficient to fully explore and describe it, collaboration and dialogue between them may enhance both, without eliminating each other's unique perspective and practice.
>
> (Yovell *et al*., 2015, p. 1526)

Moreover, since they study the same entity "understanding how the brain functions and malfunctions . . . may . . . enhance and improve both our psychoanalytic technique and our psychoanalytic theories" (Yovell *et al*, 2015, p. 1528).

It may be seen that what is claimed here and needs to be substantiated is (a) that indeed the same entity is being studied by psychoanalysis and neuroscience and (b) that as a consequence of this, understanding brain functions enhance psychoanalytic technique and theory.

In terms of support produced for these claims: The first seems to be taken for granted. The reader is simply informed that the idea that there is a 'mindbrain' is a respectable position and that neuroscientific study is the 'objective' parallel to the 'subjective' study of psychoanalysis. To this is added extensive critical comments on the possibility of thinking otherwise, with appeal to some basic intuitions. Clearly, they argue, one cannot separate the brain from the mind; what happens on the level of the mind, our thoughts, memories, feelings, obviously impacts the brain and would not be possible without it. Empirical research supports the fact that there are indeed clear mutual influences.

Yovell, Solms, and Fotopoulou seem to be saying that this dependent relationship between the mind and the brain implies that they are one and the same entity.

Yovell, Solms, and Fotopoulou's second claim, namely, that since both psychoanalysis and neuroscience address the same entity, understanding brain functions is helpful to psychoanalytic theory and practice does not describe a logical necessity. There are numerous approaches that relate even to the narrower category of mind, but they don't claim necessity in regard to them.

It is here that Yovell, Solms, and Fotopoulou introduce another source of evidence (their second argument). They present a clinical case in order to show, in practice, how neuroscience contributes to psychoanalysis.

2) The clinical illustration: How the application of neuroscientific findings may be helpful to therapeutic work

They write:

> We present this case illustration in order to demonstrate in principle our view that some neuroscientific findings may – and should – be used at times to inform and enrich the ways in which analysts work with their patients.
>
> (Yovell *et al.*, 2015, p. 1538)

While they claim that a full discussion of the implications of neuroscience for analytic practice is beyond the scope of their paper, we think that it is fair to regard this illustration, which occupies the largest section of the paper, as the authors' best shot at showing the relevance of neuroscience to clinical analytic practice. It is significant that their exemplary case is one of twice-a-week psychoanalytic psychotherapy (a point that we shall later return to).

In terms of the contribution of neuroscience, what is presented as most central to the case described is the concern of the patient, Ms A, that she was repeatedly abused as a child by the adolescent son of friends of the family. During the first year in therapy "she communicated isolated memory fragments" of this to the therapist. Her mother and the mother of the adolescent boy denied that abuse took place; her brother remembered that there was abuse, but was unsure of the details. Ms A herself was in doubt regarding the veracity of her account, at times blaming herself for making up a story in order to excuse her difficulties with men. Ms A affirmed her belief that if the events actually happened she would have been able to recall them in detail. Yovell, Solms, and Fotopoulou then provide the following very crucial few lines:

> Indeed, after a few weeks in treatment she revealed that one of her main reasons for entering psychoanalytic therapy was her hope and expectation that more detailed and complete memories of the events would emerge, concomitant with the lifting of her repression in the course of the treatment. Should this not occur, she would be inclined to believe that the abuse never took place. She felt that it was essential for her to know what had actually happened to her, and was not prepared to accept the view that how she remembered and understood her past, and what meanings she ascribed to it and to the vicissitudes of her relationship with the analyst, were just as important as finding out whether she had actually been abused. She stressed that a lot lay in the balance for her – on an interpersonal level, her relationship with her mother who flatly denied the abuse, and on an intrapsychic level, her extremely negative view of herself as someone who propagates false accusations in order to explain and justify her emotional difficulties. Ms A expected psychotherapy to provide her with keys to her missing memories which, she believed, might still lie within her, waiting to emerge.... no new memories of her past emerged during her first two years in treatment, and Ms A saw this as growing proof of the fallacy of her accusations of abuse, and as evidence of her inherent "badness".
>
> (Yovell *et al.*, 2015, p. 1532)

These lines are crucial to the clinical dilemma which Yovell, Solms, and Fotopoulou go on to highlight and in regard to which, they believe, neuroscience can contribute. The dilemma, as they see it, is how should the analyst understand and respond to Ms A's view that psychotherapy will enable her to rediscover her memories of the abuse, or establish that it never took place.

Yovell, Solms, and Fotopoulou precede their response to this question by stating that "it would seem reasonable that during the course of Ms A's therapy, some previously repressed memories of her putative abuse might come to light, as she herself and her analyst had expected". This statement provides a basis for them to rephrase the clinical dilemma. It is not how to understand Ms A's view of psychotherapy and memory, but how to understand the fact that memories have not surfaced. To this modified question Yovell, Solms, and Fotopoulou suggest that there are two common psychoanalytic responses: (a) insufficient or poor analytic work has been done or (b) "who cares?" Transference interpretation rather than reconstruction is what's important. Yovell, Solms, and Fotopoulou considers both these alternatives to be faulty, the one inviting endless analysis in search of dynamically repressed memories and the other abandoning this quest despite its real importance to victims of abuse like Ms A.

It is here that the significant contribution of neuroscience comes into play. What neuroscience shows, according to Yovell, Solms, and Fotopoulou, is that under certain stressful conditions events can fail to be encoded in declarative form. This, they write, "is significant and meaningful analytically". A third clinical option now comes to the fore; "a new avenue opens up in therapy" (2015, p. 1537). With the knowledge that it may be the case that the patient's failure to remember is not the result of dynamic reasons at all, just biological ones, the analyst can relieve the patient's distress by informing her of this. In other words, the authors' claim is that the significant contribution that neuroscience makes to analytic technique is that it recommends that at times the analyst should provide the patient with general neuroscientific data – in this case regarding memory. They write:

> Telling her that she *might* never be able to remember her abuse although she is still affected by it, and explaining to her why this might be the case, as Ms A's analyst ultimately did, relieved her anxiety and her sense of shame and guilt to a significant degree. More importantly, it freed her and the analyst to pursue the complicated task of coming to terms with her damaged internal past so that she could stop repeating it.[2]
>
> (Yovell *et al*., 2015, p. 1537)

3) Neuroscientific findings provide experimental evidence that can validate or reject analytic theories

This third step of Yovell, Solms, and Fotopoulou's presentation comes towards the end of the paper and includes one example:

[2]We should add here that the suggestion that the analyst should provide the patient with neuroscientific findings is consistent with a stance that Solms presented in a debate with Blass in 2008. Pressed by the audience to offer an example of the implications of neuroscience within the analytic setting, Solms responded that if a patient claimed to recall his birth we could tell him that based on the neuroscience of memory this could not be the case.

> There is now abundant evidence that an all-purpose *appetitive* drive similar to Freud's 'libido' does exist in the human brain … However, we also know from this same body of research that a specific drive to *attachment* exists in the human brain, which from the outset functions almost independently of the appetitive drive. The attachment drive follows different anatomical pathways, is activated by and activates different neurochemistries, is controlled by different genes, and facilitates different behaviors … This means that the forces (both constitutional and environmental) that shape our social bonds and responses to separations can and do, from the outset, follow many different and unrelated vicissitudes to those of our libidinal appetites and purposes. This conclusion strongly contradicts some important aspects of the Freudian theory of infantile sexuality.… Knowing this, an analyst would be likely to listen to his or her patients differently, and be open to attributing different meanings to some central aspects of their life stories and experiences, compared to those that he or she would have been inclined to attribute under the old theory.
>
> (Yovell *et al*., 2015, p. 1540)

4) Without neuropsychoanalysis psychoanalysis would be more scientifically isolated

This claim is made just in a line or two of the very last pages. Basically, the argument is that if psychoanalysis does not engage with neuroscience, which is the main scientific thinking about the mind, then it would be marginalized in the academic world, in research contexts and in public health funding.

How neuropsychoanalysis remains irrelevant and harmful: The misunderstanding of our critique

These arguments in support of the value of neuroscience to psychoanalysis are unconvincing and for the very same reasons that we explained in our previous articles on this topic.

1) Response to the theoretical argument

In brief, we acknowledge (as we did in our 2007 paper too) that there is an inherent tie between mind and brain. The dependence of the two is well known to anyone who ever drank a glass of wine – this isn't an advanced philosophical or scientific development based on the latest technology. But this does not mean that psychoanalysis that deals with the mind and neuroscience which deals with the brain are studying the same entity – the one subjective and the other objective. The music played by a piano cannot be separated from the piano that plays it, but we cannot conclude from this that the study of music and the study of the structure of pianos are in reference to the same entity. In contrast, the study of rhythm and the study of melody may both refer to the same musical entity. What Yovell, Solms, and Fotopoulou need to provide here to support their claim is a demonstration that the mind and the brain have an inherent connection of this latter kind. They fail to do so.

To take another example a painting could be studied in terms of its artistic value or in terms of the chemistry of which it is composed. Without the chemicals that make up the paint there would be no painting, but this doesn't mean that the artist and the chemist are studying "the same entity" in any significant sense, that the artist needs to learn chemistry or that what the chemistry professor has to say about the compound involved in the paint is relevant to the concerns of the artist. Similarly, the analyst concerned with a meaningful process of understanding the patient's inner psychical dynamics has no need to be concerned with the biological matter of the brain; studying its biology would not contribute to the understanding of the person's dynamics as they unfold in the analytic setting and it is not at all clear what it means to say that the two concerns refer to the same entity.

In line with this, of special interest is Freud's response to his imaginary, naïve interlocutor of *The Question of Lay Analysis* (1926) who asks: "What do you mean by the 'mental apparatus'? and what, may I ask, is it constructed of?" (p. 194). In direct opposition to Yovell *et al.*'s (2015) claim that "Freud always realized" that the domains of "neurobiology and psychoanalysis … are inextricable, for the simple reason that *the analytical model of the mental apparatus is embodied*" (p. 1541), Freud responds:

> It will soon be clear what the mental apparatus is; but I must beg you not to ask what material it is constructed of. That is not a subject of psychological interest. Psychology can be as indifferent to it as, for instance, optics can be to the question of whether the walls of a telescope are made of metal or cardboard. We shall leave entirely on one side the *material* line of approach.
>
> (Freud, 1926, p. 194)[3]

[3]It should be stressed here that it could be the case that two very different disciplines could meaningfully contribute to each other. This would depend not on sharing the same subject of study in the sense of observing the same physical entity, but of sharing the same aim, the same research question. A historian and a physicist could share the question of when a fire occurred in ancient times. The historian may be reading ancient scrolls and the physicist may be studying physical abnormalities of rocks. The physicist, however, cannot inform the historian of the meaning of the scroll, the ideas that it conveys (unless it was a scroll on physics).

Here it is important to note that there is a fundamental way in which the analyst seeks meaning that differs from the way the historian does, which makes some other disciplines inherently irrelevant to analysis, where they may be relevant to the historian seeking meaning. The analyst does not seek merely to understand the patient's meanings; he seeks for these to unfold in the analytic setting. So while the historian may rely on the physicist to date the scroll and this might be relevant to the understanding of its meaning, the analyst does not actively collect all available information (e.g., through home movies, school records, interviews with family members, surveillance) in order to figure out what the patient means. Gathering such external information would interfere with the analytic process of discovering meaning.

In any case, even the historian need not learn about physics or develop a dialogue with the physicist, as it is not physics per se that contributes to the historian in understanding the scroll by providing its date. Similarly, given the kind of information that neuroscience could provide, there is no more need for the psychoanalyst to be in dialogue with the neuroscientist than it is for him to be in touch with other practitioners involved in collecting data about the patient's life (e.g., a private detective).

In this context itis important to note that we have never denied relevance in the other direction, namely, of psychoanalytic understanding to neuroscientific research (counter to Bernardi's claim, 2015, p. 744). It is relevant because at times neuroscience does not only describe and explain things on the biological level but tries to connect between biology and psychology. The relevance of psychoanalysis to neuroscience is however, another matter unrelated to the one we are discussing in the present paper and unproblematic from our perspective. Depending on the kinds of statements they wish to make we agree that neuroscientists should study psychoanalysis.

In our previous papers we not only made this general theoretical argument, but demonstrated this through a study of several well-known neuropsychoanalytic papers. In Yovell, Solms, and Fotopoulou's paper it is apparent that not only is the evidence that we put forth on this matter ignored, but also that the arguments that we put forth are distorted beyond recognition. This then allows them to dismiss them outright. For example, Yovell, Solms, and Fotopoulou ascribe to us the view that:

> the domain of psychoanalysis and the domain of the neurosciences are in effect mutually exclusive: Where there is meaning, the neurosciences can have no say; where biological influences on the mind are clear and inescapable (as in brain damage or disease), psychoanalysis can have no say, and the psychological effects of such brain lesions are inherently devoid of psychic meaning (2007, p. 35). This is a radical claim, which, if taken seriously, has far-reaching ramifications for psychoanalysis. For example, it implies that psychoanalysis can cast no light on the content and meanings of the ramblings of a drunkard – since the intoxicated state has an 'organic' chemical basis. Still, this is a central claim in the current debate, and it therefore deserves examination and discussion.
>
> (Yovell *et al.*, 2015, p. 1525)

They repeat this claim over and over again throughout the paper with very slight variations (e.g., in their discussion of hardware and software, of brain illness vs meaning). They maintain that we set up an untenable division between mind and brain so that wherever the brain is recognizably involved psychoanalysis can have no say; but of course, they go on, the brain is recognizably involved in so many states and increasingly so, and in any case it is always present even when its specific role is not recognized. So that given this division that we have established (according to Yovell, Solms, and Fotopoulou), if we acknowledge the involvement of the brain, psychoanalysis can have no say at all. Since we seem to think psychoanalysis has what to say, we must think sometimes the brain is not involved – this would be a grave error (2015, p. 1515). Alternatively, we may acknowledge that the brain is always involved (and Yovell, Solms, and Fotopoulou can't deny that we openly adopt this view); but then, given the division we allegedly rely on, this should leave no room for psychoanalytic understanding – so we are gravely incoherent. Yovell, Solms, and Fotopoulou can then present their obvious solution: Drop the division and allow for the mutual involvement of mind and brain, of analytic and neuroscientific perspectives, of subjective and objective. It sounds so reasonable, so respectful, so open and friendly. Only close-mindedness could, it would seem, stand in the way of such a solution.

The problem is that *we do not at all accept the division that Yovell, Solms, and Fotopoulou ascribe to us*. On the matter cited above what we actually say (in Blass and Carmeli, 2007, p. 35 which Yovell *et al.* refer to) is that "determining the physical correlates of phenomena … does not further the understanding of the purely mental, psychological level of the mind relevant to the analytic process per se". In other words, we do not divide between situations in which the brain is involved and those in which it is not; but *we think that the different disciplines are addressing fundamentally different questions and* *are geared towards a fundamentally different process of understanding so that neuroscientific understanding is irrelevant to the unfolding of meaning in the analytic process*. Psychoanalysis, in contrast to neuroscience, aims to shed light on the meanings of whatever people, in so far as they are meaning-creating beings, express. In so far as the drunkard is still a meaning-creating-being psychoanalysis can shed light on his ramblings. At the same time neuroscientific understanding is always possible and not without value. Applied to the drunkard or to any person for that matter, it could, as we wrote in 2007, "influence action and experience by introducing biological changes. But biological explanation will not deepen our understanding of the influence of latent meanings or psychic truths – the domain of psychoanalysis" (ibid.).

Since this blatant misunderstanding of our positon on the relationship between mind and brain lies at the foundation of Yovell, Solms, and Fotopoulou's critique and is repeated throughout their paper we'd like to further clarify: (a) that we maintain that the choice to understand psychoanalytically or through neuroscientific knowledge does not depend on the state of the mind (presence of meaning) or of the brain (dominant neurological influences). It depends, rather on what the person who comes to understand is seeking; for instance, whether he is seeking to understand how the other is thinking or what brain functions are being activated. For example, instead of carrying on the present investigation of ideas in support of neuropsychoanalysis we could examine the neurology of the brains of its supporters and with the advance of technology we could hope at some point in the future to change these ideas through manipulation of their brains. While possible, to many this may seem out of place; (b) we recognize that there are situations in which two kinds of treatment may be sought simultaneously, both analytic and neurological (as in the case of analytic patients on medication). This, however, does not speak to a relationship between the treatments. In analogy, someone wishing to quit smoking may go to both hypnotic and aversive treatments, but this does not imply an inherent tie between the two; (c) we recognize that there may be situations in which psychoanalytic understanding cannot be applied. These are extreme instances in which the person can no longer be regarded as being capable of meaningful expression. We hold, and we assume Yovell, Solms, and Fotopoulou would agree, that there may be certain neuronal abnormalities, kinds of brain damage, which make analytic understanding minimally relevant. While we are speaking here of extreme cases that we imagine a typical analyst would be able to identify on his own, in our effort to think generously about the possible contribution of the neuroscientist to the analyst we suggested that "neuroscientific findings can help demarcate the limits of psychoanalysis" (Blass and Carmeli, 2007, p. 21); (d) like this last remark, our comment that "psychoanalytic models must not, of course, contradict the new cognitive findings" (2007, p. 32), does not, as Yovell, Solms, and Fotopoulou repeatedly affirm, ascribe a significant role to neuroscience, one which stands in contradiction to our explicit stance and which wrongly sets up neuroscientific findings "as the final court of appeal for psychoanalytic models". Firstly, we speak of cognitive findings, not neuroscientific ones. Secondly, non-contradiction is a very limited relationship. To illustrate: It should be clear, we think, that psychoanalytic models should also not contradict established mathematical theorems. But this does not imply that the theorems are infallible and are relevant to psychoanalysis, or that psychoanalysts should be in dialogue with mathematicians. It means that psychoanalysis is not in a position to questions mathematics; it is a legitimate domain in and of itself. We take a similar stance towards neuroscience.

In sum, Yovell, Solms, and Fotopoulou's *theoretical argument* (point 1 above) for the necessary relevance of neuroscience to psychoanalysis is completely inadequate. The fact that both psychoanalytic and neuroscientific understanding can almost always be applied to the study of the same person does not mean that their simultaneous application is mutually beneficial; more specifically, it does not mean that knowledge regarding neurons sheds light on the understanding of personal meaning. We've argued that it doesn't shed such light and that Yovell, Solms, and Fotopoulou have not shown that it does.

2) Response to the clinical illustration

Yovell, Solms, and Fotopoulou's effort to demonstrate in practice neuroscience's contribution to psychoanalysis through a *clinical illustration* also fails and in many ways. In what follows we mention five of these:

a) *The shift from personal meaning to general models of cognitive functioning:*

At first the clinical dilemma that Yovell, Solms, and Fotopoulou present is how to understand Ms A's view that psychotherapy will enable her to either rediscover her memories of the abuse or establish that it never took place – a worthy analytic question. But as we described earlier, the dilemma quickly shifts from understanding Ms A's views to understanding why memories have not surfaced. In this shift, that goes unmentioned by Yovell, Solms, and Fotopoulou, they transform the question of personal meaning to a theoretical question regarding cognitive functioning. In effect, they respond to the question of meaning with unrelated comments on general cognitive functioning. This is equivalent to raising the question of why a patient feels that his mother wanted to poison him and responding with an answer on the chemical effects of poison. The irrelevance of neuroscience to the psychoanalytic aim of understanding the patient is blatant.

b) *The theoretical contribution is not a neuroscientific one:*

The claim that under stressful conditions events may not be registered in memory in a way that could be recalled is not a statement that could be made on the basis of neuroscientific research. It is a general psychological statement that would have to be based on studies of stressful states and recall and does not require the examination of the brain. And indeed on the basis of such psychological studies the connection between stress and failure to recall has been widely accepted for many decades. Moreover, it is well known (and we would expect that Yovell, Solms, and Fotopoulou would be aware of this) that in his study of trauma and seduction Freud did not maintain that it was repression alone that stood in the way of explicit recall.

Rather repression was at times regarded by Freud as a response to the resurfacing of some kind of non-conscious remnants of events, which by their very nature (e.g., their early sexual nature) could not have been registered in a way that would allow them to be fully recalled. According to Freud such events may be regarded as "thoughts which never came about" (Freud, 1893, p. 300).[4] In light of this, what neuroscience adds to this familiar psychological finding is only information regarding its biological correlates.

We had brought this up in our previous critique and explained at length why, by definition, such a statement on memory could not be a neuroscientific one. This apparently did not go unnoticed by Yovell, Solms, and Fotopoulou, as evidenced by the fact that they mention this in a footnote (p. 1515). But instead of admitting that indeed the contribution they are referring to is not essentially neuroscientific and conceding the point, they simply add in their footnote that, nevertheless, it was "researchers who were familiar with both the psychoanalytic and the neuroscientific aspects of this controversy" who were among those who conducted a study on explicit memory of frightening events. Clearly, anecdotal information regarding the professional identity of the researchers of one study concerning the absence of explicit memory does not change the fact that the main relevant finding is not a neuroscientific one.

c) *Providing the patient with scientific information is not an analytic intervention:*

In discussing the case of Ms A it is important to stress that it is presented as a prime illustration of how neuroscience contributes to the clinical analytic process *per se*. The fact that for their prime illustration of an analytic process Yovell, Solms, and Fotopoulou chose a twice-a-week psychotherapy case may say something about their notion of what a psychoanalytic process is, e.g., that it does not refer to a unique kind of intensive process but rather can be included in a more general category of psychotherapy. More telling is the kind of intervention they advocate, namely, providing the patient with scientific information regarding memory. They found this valuable because it relieved her distressing feelings having to do with the absence of memory. With that aside she was better able to focus on dealing with her damaged internal past (Yovell *et al*, 2015, p. 1537). But is this an analytic intervention?

We think not, and the fact that this intervention may have positive consequences does not mean that it is. In fact, the idea that significant doubts

[4]Similarly in his paper on "Screen memories," he questions whether "we have any memories at all *from* our childhood: memories *relating* to our childhood may be all that we possess" (1899, p. 322). It should be noted here that Freud was quite clear that the possibility of recalling events is severely limited, whether or not they were registered in a way that would allow for this. In *The interpretation of dreams* he affirms at one point that early experiences of childhood are " 'not obtainable any longer as such,' but were replaced in analysis by 'transferences' and dreams" (1900, p. 184); and in his study of the Wolf Man he explains that "... so far as my experience hitherto goes, these scenes from infancy are not reproduced during the treatment as recollections, they are the products of construction" (1918, pp. 49–50). More generally, Freud's thoughts on the role of "repetition" in the transference (1914) and "construction in analysis" (1937) express his awareness to the fact that recollection cannot always be expected.

and fears could and should be put aside by means of the analyst providing factual information which could be regarded as neutral, devoid of interests and latent motivations, is in itself a non-analytic stance. One could think otherwise only by redefining psychoanalysis. Moreover, in this case the factual information was of a kind that specifically tells the patient that the dynamic meanings may not be involved in her predicament (her lack of memory may be due to a biological state).

To better grasp the problem with their approach one need only consider the very broad range of information that patients think are important for them to know: the chances of improving, the course that depression commonly takes, whether one's parents really love them, whether it is good to fall in love, whether one's sexual practices are normal, can the analyst contain them, etc. The analyst could opt to provide factual information and statistics regarding general tendencies on all these matters and many others. This information-providing approach downplays the importance of neuroscience – as an analyst who adopts it would need information from *all* fields that are tied to the patient's anxieties (e.g., sociology, theology, philosophy, nutrition), not necessarily neuroscience. That is, the idea that we should calm the patient's anxieties about their predicament and the treatment so that we can then get on with the analytic work is patently unanalytical.[5] To quickly highlight this point, it may be seen that if the analyst takes the information-providing approach seriously what he should actually do is give the patient a few tips on how to make better use of the Google search-engine. In this way all the helpful information that the patient needs "to get on with the analysis" would be more directly available to the patient. In the case of Ms A she would be able to simply look up the relevant info on explicit memory on her own and thus have more time left for dealing with her "damaged internal past"[6] (Yovell *et al*, 2015, p. 1537). Would Yovell, Solms, and Fotopoulou have us believe that this too would be an analytic intervention?

It should be added that were Yovell, Solms, and Fotopoulou willing to settle for the much weaker claim that they are only indirectly facilitating an analytic process, they could here concede that their intervention is indeed a non-analytic one, but a non-analytic one which, like medication, facilitates the analytic process, without being integral to it. But there would be a problem with this for several reasons: The stronger nature of their claim blocks this avenue and, as noted just above, the intervention is not based on neuroscientific information, but rather on familiar psychological insights. In addition, when medication is offered it is not usually the treating analyst who administers it. To take this avenue Yovell, Solms, and Fotopoulou must claim that it is possible to be both the analyst and an actively benevolent figure who provides neutral scientific data in order to relieve distress. Here too, in our view, this implies a non-analytic understanding of the treatment process.

[5]This is not to deny the fact that at times in life factual information is helpful.

[6]This would also spare the analyst the need to update Ms A if new and contradictory findings to emerge.

At one point Yovell, Solms, and Fotopoulou liken the neuroscientific findings regarding memory to the Oedipus complex and to the notion of projective identification – all provide models with which to understand the analytic patient. Here too one sees that the authors rely on a view of what constitutes a psychoanalytically relevant model that is so broad that it is simply untenable. Psychoanalytic models, such as that of the Oedipus complex and projective identification, relate to underlying dynamic motivations that are relevant to a meaningful understanding of the patient and to the integration of meaning. They are frames of references which make it possible to understand the clinical data in meaningful ways. Yovell, Solms, and Fotopoulou would like to include among analytic models those that precisely point to the *absence* of dynamic motivations and meaning. They tell the patient not to worry about personal meaning and motivation – a biological process may account for what's happening. To liken their biological model of memory to the Oedipus complex would be like saying that a biological theory on the effect of calorie intake on weight gain is as psychoanalytically relevant to the understanding of a patient's life-threatening obesity as are theories on phantasies of incorporation and pregnancy or on the death instinct. Indeed all offer models with which to understand the patient, but some kinds of understanding are simply irrelevant psychoanalytically.

d) *Confusion between information and consolation:*

Yovell, Solms, and Fotopoulou tell us that the information that they provided regarding a model of explicit memory calmed Ms A, relieving her anxiety about the abuse. But why should it have had this effect? According to their model it *could* be the case that the abuse occurred and because of her state of mind at the time of its occurrence she didn't form explicit memories of it. It also *could* be the case that she doesn't remember because it didn't occur. All that Ms A could learn from the information provided is that the fact that she doesn't remember everything clearly doesn't mean that it *didn't* happen. This may be the source of some relief, in the sense that she no longer need await the appearance of explicit memories for her idea that she was abused to be proven true. But the information does not tell her that it did occur. It couldn't assure her of the reality of the abuse. That kind of relief must have another source.

Here one may suggest that what is having the effect is the fact that the analyst conveys to the patient that he believes in the reality of the abuse. Yovell, Solms, and Fotopoulou make repeated reference to the 'missing memories' of 'her abuse' that 'still affected her', not to her feeling or thought that she may have been abused. They speak as though they know that it did happen; the unlikely alternative to their 'neuroscientific' explanation being that the 'memories' are still repressed. This would mean that it was confirmation and consolation rather than information regarding neuroscientific or even psychological findings on memory that impacted Ms A's state of mind. Moreover, the analyst's own belief that the abuse occurred could not be based on these findings either. They would only allow him to say 'could be'.

e) *Minimizing the role of interpretation:*

The way in which Yovell, Solms, and Fotopoulou provide information and the considerations that they bring into play in their decision to do so minimize the role of interpretation. As we have seen, while they at first raise the question of how to understand Ms A's view of psychotherapy and memory, they immediately transform it into a question of how to understand the general cognitive fact that memories haven't surfaced. They do not ever consider the meaning of the theories Ms A has developed regarding memory, repression, therapy, consciousness. But from an analytic perspective the theories she holds and how she brings them to the clinical setting are (beyond whatever external, factual sources they may rely on) expressions of inner dynamics, of her phantasies (possibly not only regarding abuse and sexuality but also about thinking, knowledge, doubt and conviction, and about the analyst as omnipotent or as judgmental). As such they are worthy of interpretation like anything else. Failing to interpret these not only limits the analytic process, it negatively impacts it. When information and consolation are introduced into the analytic setting to calm latent anxieties to the neglect of the dynamic sources of the anxieties, the dynamic sources remain untouched. In the case of Ms A, she may continue to feel that she should remember all past events and the analyst's intervention would be interpreted in that light (e.g., that he fears her demand for knowledge and thus tries to calm it in all ways possible). In contrast, interpretation of the dynamics would not only allow for their understanding, but would also modify her perceived need for concrete knowledge which emerges from the dynamics.[7] While a concern with information does not necessarily come directly at the expense of interpretation, it is striking that Yovell, Solms, and Fotopoulou's concern with it does.

3) Response to the validation of analytic theory argument

In our 2007 paper we had directly addressed the argument of Yovell, Solms, and Fotopoulou's third point, namely that neuroscience provides experimental evidence that can validate or reject analytic theories. Yovell, Solms, and Fotopoulou simply ignore us and repeat the same faulty ideas we subjected to critique. They refer to Freud's drive theory and explain how neuroscience, in line with Freud's thinking, confirms the existence of an all-purpose appetitive drive, but counter to Freud limits its scope, places an attachment drive outside this scope. Our critique was that "*Neuroscientific findings regarding motivation reveal the biological substrate of motivations, not their psychological structure*" (p. 27). Neuroscience doesn't discover an appetitive drive or an attachment drive. Rather there is observed appetitive or attachment like behaviour and neuroscience can describe what underlies this behaviour on the level of the biology of the brain. Were there no behaviour or feeling described as attachment-like there would be no possibility

[7]As noted earlier, clearly outside of the analytic setting we often relieve anxiety through information and consolation. However, because of the unique characteristics and aims of analysis to do so within analysis is a very different thing.

of neuroscience finding its centre. Therefore the question, we argued, is whether knowing about the biological correlates of motivation in any way furthers our understanding of the psychology of motivation itself. Since it just follows upon the psychological findings it is not clear how it could be an independent source of information.

It would seem that on the one hand Yovell, Solms, and Fotopoulou agree, and acknowledge that the biological level is not a crucial one; that one's views on motivation could be shaped by psychology alone (e.g., by "studying Bowlby's attachment theory and the related empirical findings of developmental psychology"). On the other hand, they argue that one can look at the biology, for example at anatomical pathways associated with attachment, and see the independence of attachment and appetitive drives. But here lies the fallacy that we pointed to in our 2007 paper. It is only when one takes what we referred to as a 'biologistic' stance and assumes that the biological substrate of motivations reveals their true and basic structure that one could conclude this. We explain:

Psychology attempts to construct theories of motivation by studying manifestations of motives and attempting to meaningfully organize them in terms of psychological considerations and categories. People express a huge number of specific motives (e.g., to get home on time, to succeed in an exam, to receive a hug, to go shopping) and many readily generalized ones (e.g., to have a family, to earn money, to succeed at work, to be loved, to avoid criticism). Psychological theories reveal and highlight less apparent psychological connections between the more apparent motives. The motives to succeed on an exam or to go shopping are not regarded as basic psychological motivations, which can, in and of themselves, well explain important aspects of human behavior. In contrast, to survive, to be socially accepted, to be loved, to be powerful, to gain mastery, to understand, to find meaning, as well as to satisfy instinctual drives and to be attached are a few of the motives that different psychological theories have, over the years, proposed as being more basic. A psychological theory would argue that what it proposes as the basic motives reflects insight into the guiding principles of human nature based on extensive study of it and allows for a better psychological explanation of the person's behavior, experience and psychic functioning. For example, it may be argued that we can better understand people psychologically, can better explain and predict their experience and behavior, when we consider both the motive to succeed on an exam and to go shopping in terms of a survival motive. The two motives obviously differ in many ways and may be associated with different biological centres, but the crucial factor in determining the category is the extent to which it enhances our meaningful understanding of the person – biology here is irrelevant. And similarly, while attachment and appetitive drives may involve different biological centres, this difference has no impact on the determination of their psychological connectedness.[8]

[8]After this connection is determined psychologically neuroscientists may try to find its biological correlates, but once again this would be post hoc and would not contribute to determining the connection or understanding it.

Psychoanalysis is concerned with a psychological understanding of motivation. Moreover, it has chosen to consider the person insofar as he is shaped by *certain kinds* of psychological motivations. These are determined by the specific insights into psychic functioning, behaviour and experience that psychoanalysis comes to from its specifically psychoanalytic perspective. This means that psychoanalysis' choice of its perspective, of how it understands what psychoanalysis is all about, also plays a crucial role in determining the motivations that it highlights (not the mere existence of a motivation and definitely not its biological correlates). Freud has expressed this very clearly in his 1917 paper "A difficulty in the path of psychoanalysis". He writes:

> Unintelligent opposition accuses us of one-sidedness in our estimate of the sexual instincts. 'Human beings have other interests besides sexual ones,' they say. We have not forgotten or denied this for a moment. Our one-sidedness is like that of the chemist, who traces all compounds back to the force of chemical attraction. He is not on that account denying the force of gravity; he leaves that to the physicist to deal with.
>
> (Freud, 1917, p. 138)

In contrast to Freud's stance on this matter, Yovell, Solms, and Fotopoulou not only reject the unique psychoanalytic focus on certain motivations, but adopt the biologistic stance which claims that the motivations with which psychoanalysis should be concerned are to be determined by what is found biologically.

4) Response to the argument that psychoanalysis will be marginalized

Our response to this argument is simple: that one should not distort psychoanalysis in order to belong to the mainstream. If in order to receive public funding of various sorts we have to agree that neuroscience contributes to psychoanalysis even when it does not, or we have to transform psychoanalysis into the kind of thing that could be contributed to by neuroscience, then we should forgo public funding. To receive it is wrong.

We should not avoid dialogue with other disciplines, but this dialogue should be of a very different kind. It should be directed at explaining what is unique about psychoanalytic theory and practice and how, as a consequence, psychoanalysis has much to offer others. This would entail making clear how and why psychoanalysis cannot make meaningful use of neuroscientific findings, but it would also make clear psychoanalysis' special contribution to science and mental health.

Preserving the spirit of psychoanalysis

There are many disciplines that study the person, his behavior, thought and feelings and how he evolves and changes. These include numerous psychological approaches, sociological, anthropological, historical, philosophical, theological, as well as biological and more specifically neuroscientific ones. The different disciplines have different concerns and aims and as a consequence the common denominator of 'the person' to which they refer does not in and of itself make the insights of one discipline relevant to the other. There may be a broad common subject of study, but the choice to understand in a certain way delimits that subject. Similarly, there are many different psychological approaches that study the mind. The best psychoanalytic understanding doesn't require dialogue with them all because the basic concerns of psychoanalysis differ from those of other approaches (Blass, 2015).

Yovell, Solms, and Fotopoulou propose that psychoanalysis and neuroscience share the subject of study – 'mindbrain'. It is very unclear what this entity refers to – why isn't there just a mind that stands in certain relationships to the brain? It would seem that this term begs the question; instead of exploring the question of what do each of them study in order to examine whether neuroscience is necessary for psychoanalytic understanding it simply posits (without adequate argument) this mutual object of study.[9] But in any case, even if the existence of such an entity was clear and reasonable, then too, one's choice of focus shapes and limits the subject of study.

Yovell, Solms, and Fotopoulou don't seem to appreciate this limitation, but interestingly, they can't seem to appreciate this specifically in relation to the contribution of neuroscience to psychoanalysis. They allow that the psychoanalyst could choose to be concerned with psychoanalysis without taking into account *all* other psychological approaches, even though these other approaches also are concerned with the mind. They don't argue that the thinking and findings of philosophy and sociology are *necessary* for the best psychoanalytic understanding of the individual, even though these disciplines also study the person, refer to the same 'entity', to use their terminology. Were the criterion for necessary dialogue the common denominator of being involved in the study of the mind, let alone the 'mindbrain,' psychoanalysis would not be able to take a step forward without first considering the theories and findings of all of the humanities and social sciences. But Yovell, Solms, and Fotopoulou are concerned only with the necessity of neuroscience. Only neuroscience *must* be taken into account. And it is here too that their biologism becomes apparent – their latent belief that psychology is actually grounded in biology and that it doesn't have an independent reality. In other words, their reason for maintaining that psychoanalysis has to learn from neuroscience is not really, as they claim, because the two fields deal with the same basic entity (so many other fields do so too); rather it is because of the special foundational status they ascribe to biology. As we have seen, for them biological evidence defines motivation, and more broadly provides the true evidence that can validate or reject psychological claims.

Attributing relevance to that which is irrelevant to psychoanalysis is harmful to the field. As is apparent from our discussion of the case of Ms A it distorts the very essence of analytic practice, in fact, its very spirit; it transforms psychoanalysis into some other kind of more general practice that could benefit the patient by providing him with information about bio-

[9]That is, Yovell, Solms, and Fotopoulou make use of the notion of mindbrain to explain the necessity of neuroscience to psychology and cite many authors in this context. They don't, however, adequately argue for the existence of such an entity as mindbrain or that it is psychoanalysis' subject of study.

logical findings in regard to cognition (and again strangely only with information about biological findings, rather than from all anxiety-relieving facts of life). This distortion of the very spirit of psychoanalysis as put forth by Freud cannot be made right by repeatedly professing appreciation of psychoanalysis or by quoting Freud's few comments on the potential contribution of biology, which stand in contradiction to many of his other writings and the broader spirit of his work.

The failure of Yovell, Solms, and Fotopoulou to understand the meaning and consequences of the choice to understand in a specifically psychoanalytic way leads them to regard this choice as arrogant, as an expression of unwillingness to step outside one's own limited framework in order to receive helpful knowledge from outside. As they say "There can be no excuse for deliberate, self-imposed ignorance; it is not a virtue." In turn, they present their own stance as a reliable and humble academic one, which is open to knowledge wherever it comes from.

We hope that in this paper we have succeeded to explain why this is a fundamentally mistaken and unfair presentation of the ideas we are putting forth. To choose to understand psychoanalytically is not to dismiss the value of other fields, but rather to recognize the limits of psychoanalysis and the essential ways in which other fields differ from it. We recognize that there are different ways of understanding: psychoanalytical and neuroscientific, artistic and chemical, musical and structural, historical and physical and we respect each of these ways and what it has to offer, but without blurring the distinctions between them. We see this as a humble and responsible stance and we are willing to forgo the awards and praise bestowed upon neuropsychoanalysts in order to uphold it.

In fact, we think it is important to fight for this stance and our critiques of neuropsychoanalysis have been part of this fight. They are addressed to all those who are concerned with preserving psychoanalysis as a field concerned with understanding the meaning of inner psychic dynamics and the transformation of these dynamics through such meaningful understanding as they unfold in the analytic process. This is what's at stake.

As neuropsychoanalysis establishes itself as a regular contributor to psychoanalytic conferences, programs and publications and as it competes for analytic funding, psychoanalysts can no longer simply stand by the side and enjoy the positive PR that psychoanalysis receives through its supposed dialogue with neuroscience, or the feeling of being broadminded and scientific that conducting such dialogue may afford. Given what's at stake psychoanalysts cannot just assume that there must be at least something to neuropsychoanalysis and if it doesn't help it can't harm.

What we have shown in the present critique of Yovell, Solms, and Fotopoulou's paper is that there is simply 'nothing to it' and it does harm. If their paper is the best defense that neuropsychoanalysis has to offer, then the field has absolutely no *raison d'être* and the analytic world should recognize this and respond accordingly, not merely suspend judgment in the vague belief that there could be some other neuropsychoanalysts that might better defend the field. If they are there, then they should speak out – we would welcome the dialogue.

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